

TCS349: Responsible and Explainable AI

Unit 2, Lecture 1

Principles of Responsible AI & the Trustworthy AI Lifecycle

Moving from Unit I's focus on ethics and fairness to Unit II's framework for building AI systems people can trust - from design through deployment and beyond.

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Fairness



Transparency



Accountability



AI Lifecycle

1. From Unit I to Unit II: Why Principles Now?

Unit I focused on one dimension of Responsible AI in depth: detecting, measuring and mitigating bias.

Real-world AI systems need more than fairness alone - they must also be understandable, governed and privacy-respecting.

Unit II introduces the broader principle-based framework that fairness fits inside.

Today's lecture sets up the structure and vocabulary used throughout the rest of Unit II.

2. What Is Responsible AI?

Responsible AI (RAI) is the practice of designing, building and deploying AI systems in ways that are ethical, lawful and aligned with human values.

It shifts AI development from "can we build this?" to "should we build this, and how do we build it responsibly?"

RAI is not a single technique - it is an umbrella of principles, processes and governance structures applied throughout a system's life.

Organisations increasingly formalise RAI through internal policies, review boards and regulatory compliance.

3. Core Principles of Responsible AI - Overview

Principle	One-Line Meaning
Fairness & Non-Discrimination	Equitable treatment across individuals and groups
Transparency & Explainability	Understandable reasoning and honest communication of limits
Accountability & Human Oversight	Clear responsibility and the ability to intervene
Privacy & Data Governance	Lawful, respectful use of personal data
Safety, Robustness & Reliability	Consistent, safe performance under real conditions

4. Principle: Fairness & Non-Discrimination

Ensures an AI system treats individuals and groups equitably, without unjustified discrimination.

Builds directly on the bias-detection and fairness-metric techniques studied in Unit I.

Requires ongoing subgroup monitoring, not just a one-time check before deployment.

Example: a hiring AI audited regularly for equal opportunity across gender and demographic groups.

5. Principle: Transparency & Explainability

Ensures stakeholders can understand how and why an AI system produced a particular output.

Covers both the system's internal logic (explainability) and clear communication about its capabilities and limitations.

Builds trust with users, regulators and affected individuals who may need to contest a decision.

The full depth of this principle is the subject of the next lecture.

6. Principle: Accountability & Human Oversight

Ensures a clear chain of responsibility exists for an AI system's decisions and outcomes.

Requires defined roles, escalation paths and mechanisms for humans to review, override or halt automated decisions.

Prevents a "many hands, no owner" situation where no one is responsible when something goes wrong.

Explored in depth in Lectures 13-14 through accountability frameworks and governance.

7. Principle: Privacy & Data Governance

Ensures personal and sensitive data used by an AI system is collected, stored and processed lawfully and respectfully.

Covers informed consent, data minimisation, purpose limitation and protection against misuse.

Especially critical for AI used in surveillance, healthcare and finance.

Covered in depth in Lectures 16-17 on privacy concerns and data-protection principles.

8. Principle: Safety, Robustness & Reliability

Ensures an AI system performs consistently and safely, including under unexpected or adversarial conditions.

Covers testing for edge cases, resistance to manipulation, and graceful failure when the system is uncertain.

Poor robustness can silently reintroduce bias or unfairness that earlier testing failed to catch.

Works together with the other four principles rather than standing alone.

9. What Is a Trustworthy AI Lifecycle?

A trustworthy AI lifecycle applies responsible-AI principles at every stage of a system's life, not only at launch.

It treats trust as something built continuously, from initial design through retirement of the system.

Each stage has its own responsible-AI checks and expected artefacts, such as documentation and audit logs.

The lifecycle view is what connects the five individual principles into one repeatable process.

10. Lifecycle Stage 1-2: Design & Data

Design: define the system's purpose, intended users and acceptable-use boundaries before writing any code.

Data collection: assess the representativeness, consent basis and known limitations of the training data.

Early identification of risk here prevents costly bias or privacy issues later in the lifecycle.

Responsible AI treats this stage as the most cost-effective point to intervene.

11. Lifecycle Stage 3-4: Development & Testing

Development: build the model while documenting design choices, assumptions and known trade-offs.

Testing and validation: evaluate accuracy, fairness, robustness and explainability before release, not only after.

Subgroup analysis and confusion-matrix measures from Unit I are applied here as standard checks.

Any failed check should be resolved or explicitly justified before moving to deployment.

12. Lifecycle Stage 5-6: Deployment & Monitoring

Deployment: release the system with clear documentation of its intended use and limitations.

Monitoring: continuously track performance, fairness and drift after the system is live.

Feedback loops (Unit I) make ongoing monitoring essential, since bias can emerge only after real-world use begins.

A retirement or update plan should exist for when the system no longer meets its responsible-AI standards.

13. Governance Across the Lifecycle

Governance structures - review boards, policies and audit trails - connect the principles to the lifecycle in practice.

Clear ownership at every stage ensures responsible-AI checks are actually performed, not just documented.

Regulatory compliance is treated as a minimum baseline, with organisational governance often going further.

This sets up Lectures 13-14, which examine accountability frameworks and governance in detail.